Independent Crypto Market-Microstructure Research

RESEARCH & DEVELOPMENT FUNDING BRIEF

I build multi-exchange market-microstructure infrastructure and test market and execution hypotheses under rules fixed before the results are examined, publishing every outcome — including the failures — in public.

THE PROBLEM

A recurring problem in crypto research is that hypotheses, execution costs and failure criteria are often not fixed before results are examined. Settled afterwards, a real finding is hard to distinguish from the best of many unrecorded attempts.

WHAT I HAVE BUILT

OI-VOL — multi-exchange market-microstructure research infrastructure. Venue connectors for seven exchanges, with deep collection on Binance USDT-M, analysing price, open interest, volume, funding and basis, and cross-exchange activity. Collectors run under watchdog supervision. Research so far has not established a reliable directional edge; OI-VOL is therefore positioned primarily as market-activity research infrastructure rather than a predictive signal engine. No candidate rule has been validated as predictive.

Whale Lab — a controlled study of execution state for large TWAP parent orders on Hyperliquid: which terminal state a live order reaches, when, and how much more size executes. Deliberately not a price-prediction study.

WHAT I HAVE FOUND SO FAR

Most apparent edges do not survive costs or integrity checks.

+3.78 bps

AGAINST A 58 BPS HURDLE

Statistically real is not the same as tradeable. A measured perp–spot basis effect of +3.78 bps sat against a 58 bps round-trip cost hurdle — roughly fifteen times too small to trade. The pre-committed stopping rule fired and closed that line of research.

42.4 bps

MEASURED, NOT ASSUMED

Execution cost has to be measured, not assumed. Measured directly at 42.4 bps all-in for a taker round trip, with exits costing about three times entries. From 7,149 fills in one account over January–July 2026 — that limitation is part of the finding.

94.2351%

AGAINST A 99.9% FLOOR

Integrity checks have to be able to stop a study. Whale Lab's first attempt reconciled 94.2351% of parent orders against a required 99.9% floor. It failed, and it stays failed: the data was not repaired to produce a usable result, and no model was fitted.

WHAT IS BEING TESTED NOW

Whale Lab v2. The preregistration is **frozen**. The current phase is **implementation and integrity/acceptance testing**. **No model has been fitted, and the prospective test has not started.** Confirmation will come from a single **30-day prospective window, opened only after the model is frozen** — one window, one reading. No result exists, and no profitable strategy is claimed.

WHY THIS IS DIFFERENT

The question, the cost hurdle and the conditions for declaring failure are committed before any data is examined. A failed integrity gate ends the study, and the failure is published rather than repaired. The process is designed so that a null result remains a valid, publishable outcome.

FUNDING REQUEST

PROVISIONAL FUNDING RANGE

USD 200–500

A one-off research and development runway for infrastructure, data, compute and prospective testing. It is not trading capital, and none of it funds market positions.

USE OF FUNDS

An always-on collection host would move data collection off a personal machine and reduce the sleep-related gaps already observed in the archive. Market data and archive access. Compute and storage — reprocessing runs in multi-hour passes on a single laptop, and one study alone holds about 85 GB. Development time, the prospective test window, and tooling that makes results auditable.

NEXT MILESTONES

Pass the acceptance tests › reprocess development data under the frozen rules › fit and freeze the model › run one 30-day prospective window › read once by the pre-committed rule and publish the outcome either way.

PUBLIC EVIDENCE

`github.com/viter999/quant-market-research`

Method, preregistrations, findings including failures, and a complete worked example from preregistration to null result.

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